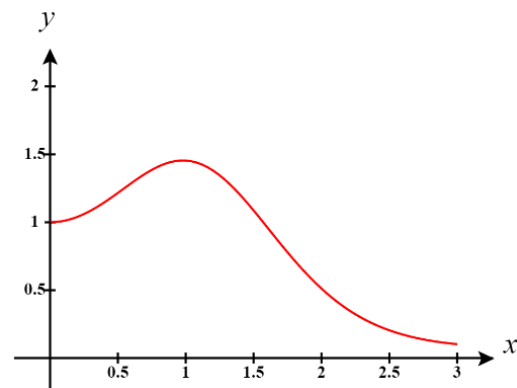
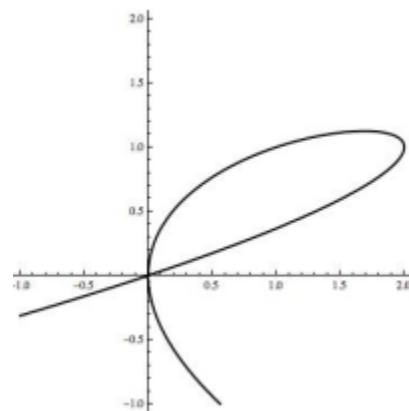


- Find the equation of the line tangent to the given curve at the point $(2,3)$. $\ln(x^2 - 3)y - yx = 6$
- True or False? The tangent line to the function $y = (x^2 + 1)^{\cos(x)}$ (shown below) is horizontal when $x = 1$.

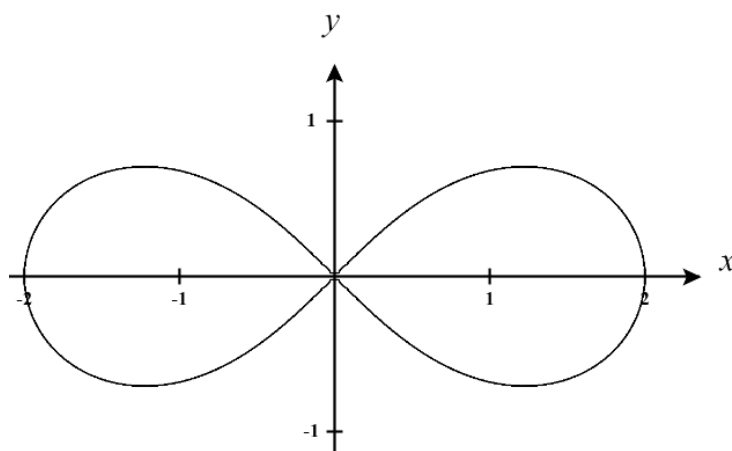


- The following is the graph of $2y^3 - 3xy + x^2 = 0$. There are two points on the graph where $y = 1$. Determine the equation of *both* tangent lines to the graph where $y = 1$.



- The lemniscate shown below is described by the equation $(x^2 + y^2)^2 = 4(x^2 - y^2)$ and is used in typography to denote infinity (∞)

- Find $\frac{dy}{dx}$ for points on the graph.
- Are there any points on the curve for which $\frac{dy}{dx}$ is undefined? If so, find all such points.
- Determine the equation for the tangent line at the point $\left(\sqrt{\frac{3}{2}}, \frac{1}{\sqrt{2}}\right)$



- Consider a circle centered at the origin with radius r .
 - If $r = 5$, show that the tangent line at the point $P(3,4)$ is perpendicular to the radius of the circle.
 - Show that for a circle of any radius that the tangent line at the point $P(x,y)$ is perpendicular to the radius of the circle.